Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · WELCH'S ANOVA

3+ groups, unequal variances

Table 1. Descriptives by group

Group	N	Mean	Std. Dev.
control	20	32.48	7.47
drug_a	20	33.84	6.91
drug_b	20	37.81	7.79

Table 2. Welch's ANOVA

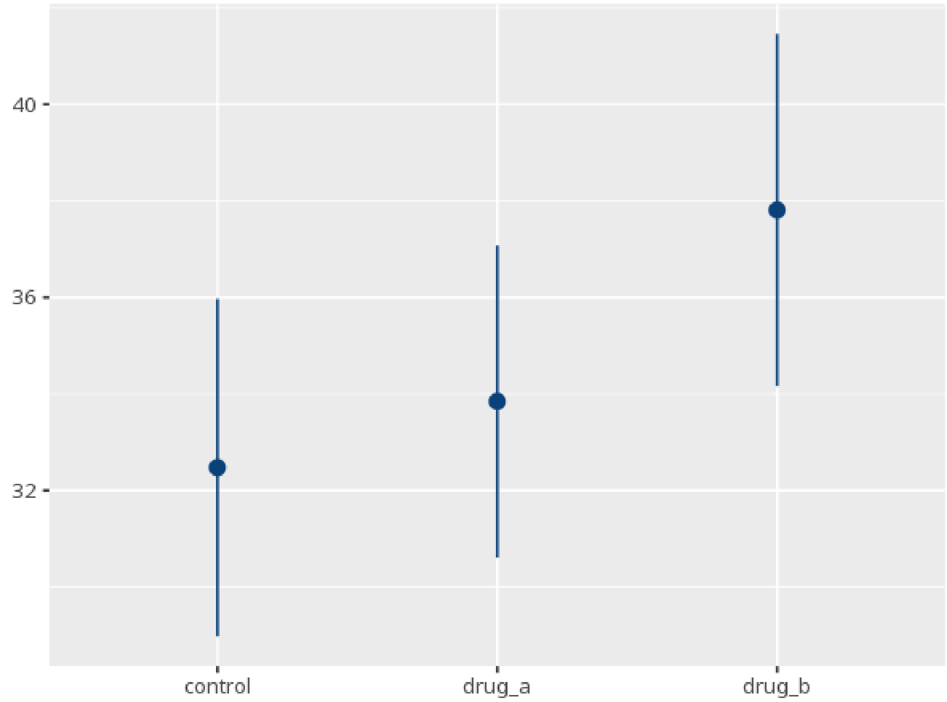
F	df1	df2	p	ω^2 [95% CI]
2.58	2	37.90	.089	0.07 [0.00, 1.00]

Table 3. Games-Howell post-hoc

Pair	M_{diff}	p_{adj}	95% CI
control - drug_a	−1.37	.820	[−6.92, 4.18]
control - drug_b	−5.33	.082	[−11.22, 0.55]
drug_a - drug_b	−3.96	.218	[−9.65, 1.72]

Welch's adjusts the degrees of freedom so equal variances are not assumed (df2 is fractional); within-group normality is still assumed and checked with Shapiro-Wilk per group. (Shapiro per group: control W=0.97, p=.688; drug_a W=0.90, p=.050; drug_b W=0.92, p=.113) — normality looks reasonable across groups

Figure. Group means



Understanding the numbers

F. The ratio of between-group to within-group variance, computed without assuming equal variances. Here, $F(2, 37.90) = 2.58$.

df1. The numerator degrees of freedom - one less than the number of groups. Here, df1 = 2.

df2. The (fractional) denominator degrees of freedom, adjusted downward when group variances are unequal. Here, df2 = 37.90.

p. The probability of an F this large if all group means were truly equal. Here, $p = .089$ - at or above the $\alpha = 0.05$ threshold.

ω². The effect size for a Welch F-test - the estimated share of variance explained by group membership, converted directly from F/df1/df2 since Welch's test has no fitted model to compute it from. Here, $\omega^2 = .07$ [.00, 1.00].

Mean. The average outcome for that group. Here, each of the 3 groups has its own mean in the descriptives table above.

N. The number of observations in that group. Here, each of the 3 groups has its own N in the descriptives table above.

Std. Dev.. How spread out outcome scores are within that group. Here, each of the 3 groups has its own SD in the descriptives table above.

M diff. The difference between two group means in the Games-Howell post-hoc table. Here, each pairwise contrast above reports its own mean difference in the Games-Howell table.

p (adj). The pairwise p-value, Games-Howell-adjusted (variance-robust, like the omnibus test) for the number of comparisons. Here, each pairwise contrast above carries its own Games-Howell-adjusted p.

95% CI. The range likely to contain the true mean difference for that pair. Here, each contrast's 95% CI is shown in the Games-Howell table above.

How to read this test

A one-way ANOVA that relaxes the equal-variance assumption but still assumes roughly normal data within each group. A significant F means at least one group mean differs from the others; the Games-Howell post-hoc (also variance-robust) then shows which specific pairs differ. Your significance threshold (α) is 0.05.

APA template: Welch's ANOVA gave $F(2,37.90)=2.58$, $p = .089$, $\omega^2=.07$ [.00, 1.00].

Statistical basis: Welch's ANOVA (heteroscedasticity-robust F, fractional df) (Welch, B. L. (1951). "On the comparison of several mean values: an alternative approach." Biometrika, 38(3-4), 330-336.); Games-Howell post-hoc comparisons (also variance-robust) (Kassambara A (2023). rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package.); ω^2 effect size (F-to-omega^2 conversion, no fitted aov model needed) (Ben-Shachar MS, Lüdecke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.